

Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent 4-space indentation followed (PEP8 / Prettier).
2	Proper File Structure	Files and folders are logically organized	Yes	Backend split into routers/ai/models; frontend split into pages/components/context/api.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. monthly_income, credit_utilization, financial_health_score.
4	Function / Method Names	Functions are descriptively named	Yes	e.g. calculate_financial_score(), generate_repayment_plan().
5	Code Comments	Inline and block comments explain logic	Yes	Comments added for score calculation and AI prompt logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate routers per feature (auth, debts, expenses, dashboard, ai, reports).
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared schemas/utilities reused across routers.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	FastAPI exception handlers and try/except around AI API calls with offline fallback.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Financial Score Engine	Python (FastAPI)	Dashboard, AI Analysis	High
2	Gemini Client	Python	AI Debt Analysis, AI Assistant	High
3	Auth Utilities (JWT/bcrypt)	Python	All protected routes	High
4	API Client (Axios instance)	JavaScript (React)	All frontend pages	High
5	Chart Components (Recharts)	React	Dashboard, Reports	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized backend/frontend project structure.
Readability	5	Clear naming and consistent style throughout.
Reusability	5	Reusable routers, services and React components.
Documentation / Comments	4	Key logic documented; README covers setup and features.
Overall Score	5	Good coding practices followed.